Lab 1

Introduction to Programming - Lab 1

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Answer the theory questions. Follow the instructions in the Practical section for the programming exercises.

Theory

Question 1: Editor and interpreter roles

An editor highlights Python keywords and reports no warnings. Does this prove that the program will run correctly? Distinguish editing support from program execution, and give an example of a mistake that syntax highlighting alone would not resolve.

Question 2: Independent conditions

Compare two independent `if` statements with an `if` followed by `elif`. Use an integer that is both positive and even to illustrate the difference. Explain when printing two messages is intentional and when it indicates an incorrectly chosen branching structure.

Practical

Submission: provide one complete Python 3 script for each assigned exercise, together with your test cases. Explain important decisions as code comments. Screenshots are not required.

Scope: use variables, scalar values, strings, arithmetic, conditions, and `for` or `while` loops. Built-in input/output, conversion, `range`, `len`, `abs`, `min`, `max`, and basic string operations are permitted. Do not define functions or classes, import libraries, or use lists, tuples, dictionaries or sets. Algorithmic reasoning is allowed in this Practical section.

Input and output: each numeric value is provided on its own line, in the order stated in the question. A word, command or string occupies one whole line. Print only the requested result, without input prompts. In compact examples, `/` marks a line break; the slash is not part of the actual input or output. Test at least one boundary case and one ordinary case in addition to the example. Multiple valid outputs are accepted only where explicitly stated.

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Question 3: Choosing a uniform letter case

Count uppercase and lowercase letters in a word. Convert the entire word to whichever case requires changing fewer letters. When the counts are equal, choose lowercase. Use a loop to obtain the two counts.

Input: Read one mixed-case word.

Output: Print the converted word.

Bounds: Word length is 1 through 100; English letters only.

Example: input `cODe`; output `code`.

Question 4: The omitted integer

Integers from 1 through n were written down, but exactly one is absent. The remaining n-1 integers are distinct and may arrive in any order. Find the missing value using numeric state; do not store the full sequence or use a set.

Input: Read n, then the n-1 remaining integers.

Output: Print the missing value.

Bounds: 2 <= n <= 10000.

Example: input `5 / 2 / 3 / 1 / 5`; output `4`.

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